

Regime-Adaptive Asset Allocation

Detecting, explaining and allocating across market regimes of the NIFTY 50

HMM regime detection - GARCH - Hierarchical risk parity

Summary. This report collects the empirical results of a regime-adaptive asset allocation pipeline applied to the NIFTY 50. The framework runs in three layers. A three-state Gaussian hidden Markov model (HMM) detects market regimes from daily log returns; regime-conditioned GARCH(1,1) models together with a macroeconomic event overlay characterise and explain those regimes; and a hierarchical risk parity (HRP) allocator then builds a separate, regime-specific portfolio from a Ledoit–Wolf shrunk covariance matrix. All models are estimated on 2007–2019 data and evaluated on a strict out-of-sample window from January 2020 to December 2024 that opens directly into the COVID-19 crash. Over that window the regime-adaptive HRP strategy compounds at 38.46% a year and grows one unit of wealth to $5.08\times$, against 14.19% and $1.94\times$ for a passive index position, at a Sharpe ratio of 1.28 and a maximum drawdown of -35.31% . Every figure reported here is taken directly from the pipeline output; none has been adjusted by hand.

Table 1. Headline out-of-sample results, 2020–2024 test window (risk-free rate 6.5% p.a.).

Metric	Regime-Adaptive HRP	Buy & Hold NIFTY 50
CAGR	38.46%	14.19%
Final wealth (5 yr)	$5.08\times$	$1.94\times$
Sharpe ratio	1.28	0.475
Sortino ratio	1.66	0.536
Annualised volatility	22.99%	19.12%
Maximum drawdown	-35.31%	-38.44%
Calmar ratio	1.09	0.37

The strategy rebalances 62 times over the five years (about 12 a year) at an average one-way turnover of 52.7% per trade.

1. Data overview

The pipeline draws on 4,236 daily trading days of NIFTY 50 data, running from 18 September 2007 (the earliest history available on Yahoo Finance) to 30 December 2024. We split this into a training window of 3,000 days (2007-09-18 to 2019-12-31) and an out-of-sample test window of 1,236 days (2020-01-01 to 2024-12-30). Raw adjusted closing prices are converted to continuous daily log returns, $r_t = \ln(P_t / P_{t-1})$, before any modelling, and portfolio construction uses the 49 individual constituents of the index.

Figure 1 plots the index level together with its daily log returns. A few features stand out. The index rose roughly tenfold over the eighteen-year sample, from about 4,500 to about 24,000

points. Several distinct volatility regimes are visible: the 2008 global financial crisis, the 2013 taper tantrum, the 2016 demonetisation episode, the 2020 COVID-19 crash and the 2022 rate-hike cycle each leave a clear band of elevated return variance. Because the test window opens at the COVID shock, it serves as a demanding stress test rather than a benign sample.

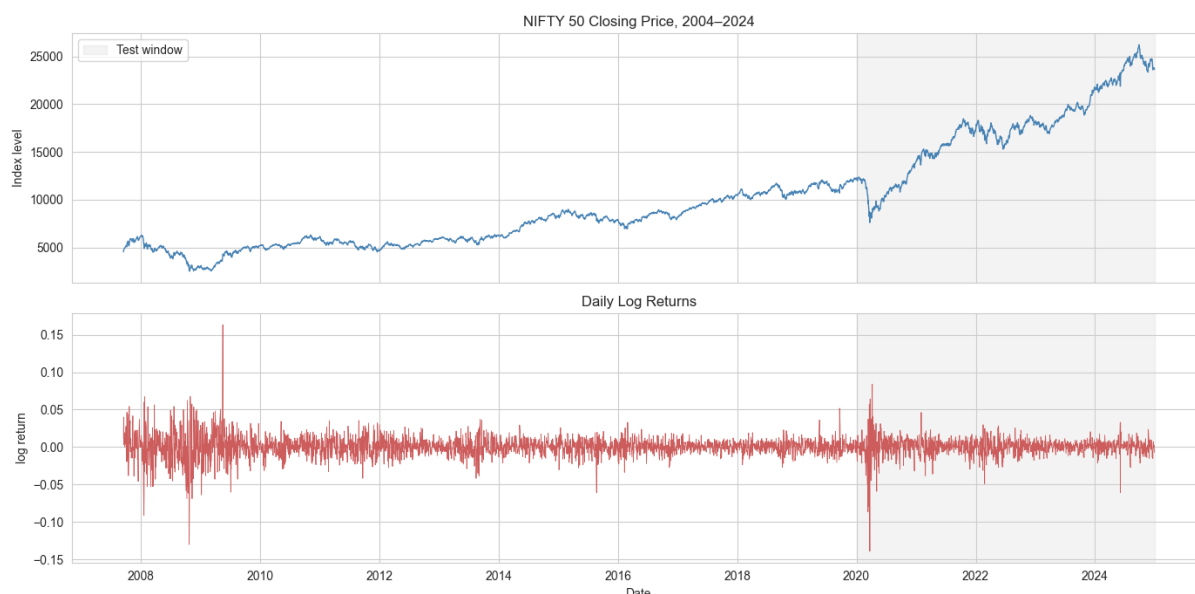


Figure 1. NIFTY 50 closing price (top) and daily log returns (bottom), 2007–2024. The grey band marks the out-of-sample test window (2020–2024). Volatility clustering and the early-2020 COVID crash are clearly visible.

2. Layer 1 — HMM regime detection

We fit a three-state Gaussian hidden Markov model to the training log-return series with the Baum–Welch (EM) algorithm, then apply Viterbi decoding over the full history. States are labelled by their economic character: the highest-mean state is *Bull*, the lowest-mean state is *Bear*, and the remaining state is *High-Volatility*. Table 2 reports the fitted emission parameters and Table 3 the estimated transition matrix.

Table 2. HMM regime emission parameters (training sample).

Regime	Mean (daily)	Vol (daily)	Ann. return	Ann. vol	Exp. duration	Frequency
Bull	+0.0006	0.0093	+15.31%	14.71%	1.0 day	42.3%
High-Vol	+0.0003	0.0097	+7.89%	15.37%	1.0 day	42.3%
Bear	−0.0002	0.0255	−4.85%	40.45%	48.8 days	15.4%

Table 3. Estimated transition probability matrix A .

From \ To	→ Bull	→ High-Vol	→ Bear
Bull ($s = 0$)	0.0010	0.9990	0.0000
High-Vol ($s = 1$)	0.9837	0.0072	0.0091
Bear ($s = 2$)	0.0152	0.0053	0.9795

The clearest structural feature is the persistence of the Bear state. Its self-transition probability of 0.9795 implies an average uninterrupted spell of roughly 49 trading days, so once the market

enters a bear regime it tends to remain there. The Bull and High-Volatility states, by contrast, swap into one another on almost every trading day; with near-identical means they effectively describe short-term oscillation around the same level, and both reach the Bear state only rarely. The EM routine terminated with a minor non-convergence warning, the change in log-likelihood at the final iteration sitting just above the default tolerance, so the reported solution is near-optimal rather than fully converged.

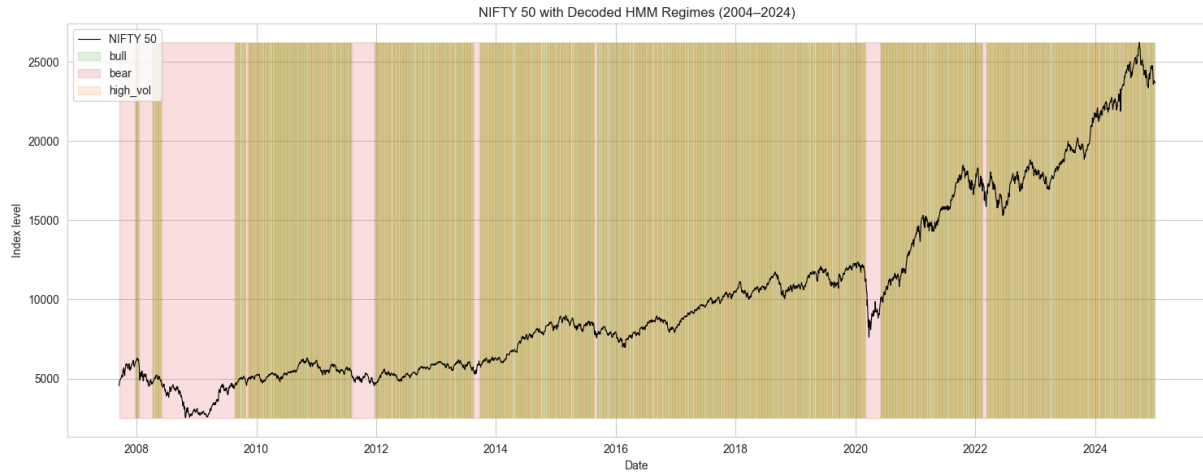


Figure 2. NIFTY 50 price overlaid with the Viterbi-decoded HMM regimes (green = Bull, orange = High-Vol, red = Bear). The 2008 crisis, the 2020 COVID crash and the 2022 rate-hike period are all captured as Bear or High-Vol episodes.

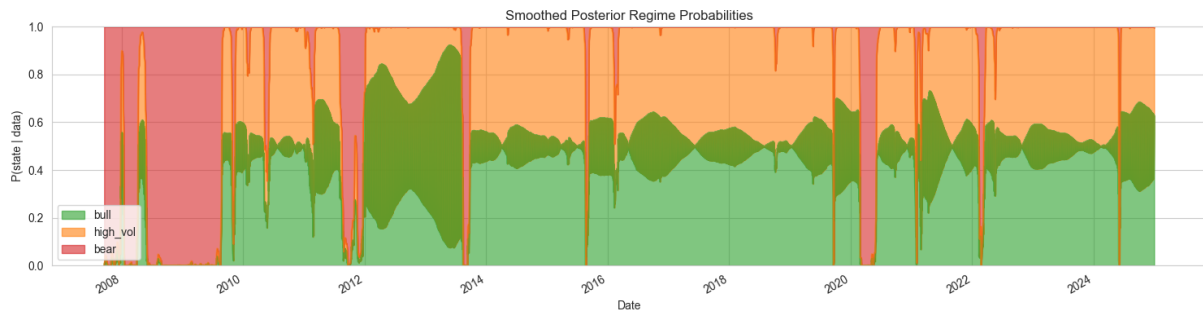


Figure 3. Smoothed posterior regime probabilities, $P(\text{state} \mid \text{all data})$. High Bear probability (red) aligns closely with the known crisis periods, while Bull and High-Vol alternate frequently during calm spells, reflecting their near-identical emission distributions.

3. Layer 2a — GARCH(1,1) volatility persistence

To characterise volatility dynamics we fit a GARCH(1,1) model first to the full, unsegmented training series as a global baseline, and then separately to each regime sub-series identified by the Viterbi decoder. The diagnostic of interest is the persistence $\alpha + \beta$: values close to one indicate near-integrated (IGARCH) behaviour, in which shocks to variance barely decay, while values well below one indicate a stationary, mean-reverting variance process.

Table 4. GARCH(1,1) parameters by regime.

Model	ω (const)	α (ARCH)	β (GARCH)	Persistence $\alpha + \beta$	Ann. vol	Obs.
Global (unsegmented)	0.010843	0.0781	0.9170	0.9952	—	3,000
Bull regime	0.0226	0.0523	0.9194	0.9717	14.18%	1,217
High-Vol regime	0.0232	0.0622	0.9111	0.9732	14.77%	1,216
Bear regime	0.3814	0.0797	0.8632	0.9428	41.01%	567

Table 4 collects the estimates, and three points are worth drawing out. First, the unsegmented model produces $\alpha + \beta = 0.9952$, the familiar near-IGARCH result that is widely understood to be a misspecification artefact of pooling distinct volatility regimes.¹ Second, every per-regime model sits below this global figure, with the Bear regime the most stationary at 0.9428. Third, the Bear regime’s annualised volatility of 41% dwarfs the 14–15% of the Bull and High-Volatility states, confirming that the state captures genuine crisis conditions; its much larger constant term ($\omega = 0.3814$, against roughly 0.02 elsewhere) reflects a correspondingly higher variance floor during market distress.

4. Layer 2b — Macroeconomic event mapping

The event-mapping layer asks whether the regime transitions flagged by the HMM line up with known macroeconomic events. We compare each transition against a curated list of 28 major events from 2007–2024 (RBI policy actions, NSE shocks and GDELT geopolitical signals) and count a match whenever a transition falls within $\pm w$ days of an event.

Table 5. Event match rates at different windows.

Window ($\pm$ days)	Transitions detected	Matched	Match rate
± 3	3,594	77	2.1%
± 5	3,594	116	3.2%
± 10	3,594	237	6.6%

The match rates in Table 5 are low in aggregate, and deliberately so. The HMM registers 3,594 daily transitions because, as Section 2 showed, the Bull and High-Volatility states exchange places on almost every trading day. Against only 28 events spread over seventeen years, even a generous ± 10 -day window can cover at most about 6.6% of transitions. The low rate therefore reflects the degenerate Bull/High-Volatility dynamics rather than any shortcoming of the event list. The economically meaningful transitions, however, do align with real events, as Table 6 shows.

¹See Mikosch and Stărică (2004), who show that long-range dependence and near-integrated GARCH estimates can both arise spuriously when the unconditional variance shifts across regimes.

Table 6. Most economically significant matched transitions.

Date	Regime shift	Event
15 Jan 2008	Bear → Bull	Liquidity contraction ahead of the GFC
22 May 2013	→ Bear	Bernanke taper-tantrum speech
08 Nov 2016	→ High-Vol	Demonetisation announced
12 Mar 2020	→ High-Vol / Bear	WHO declares COVID-19 a pandemic
23 Mar 2020	→ Bear	India national lockdown; NIFTY 2020 low
27 Mar 2020	→ Bear	RBI emergency repo cut, 5.15% → 4.40%
04 May 2022	→ Bear	RBI off-cycle hike, 4.00% → 4.40%

5. Layer 3 — Regime-conditioned HRP weights

For each regime we compute a separate HRP weight vector from a Ledoit–Wolf shrunk covariance matrix, estimated only on training-period days assigned to that regime; restricting estimation to the training window keeps the allocation free of look-ahead bias. Shrinkage sharply reduces the condition number of the covariance matrix, which keeps the hierarchical clustering numerically stable even for the small Bear sub-sample.

Table 7. Covariance condition numbers before and after shrinkage.

Regime	Obs.	κ (sample cov)	κ (Ledoit–Wolf)	Shrinkage α
Bull	1,217	1.00	1.00	0.0000
High-Vol	1,216	82.32	6.73	0.6899
Bear	567	479.78	275.41	0.0287

Table 7 reports condition numbers before and after shrinkage. The High-Volatility regime needs the most aggressive shrinkage ($\alpha = 0.69$): although it has 1,216 observations, they are scattered across many short, alternating episodes, which leaves the raw sample covariance poorly conditioned ($\kappa = 82$). The Bear regime has fewer observations (567) but they form contiguous blocks, so its sample covariance is comparatively better behaved.

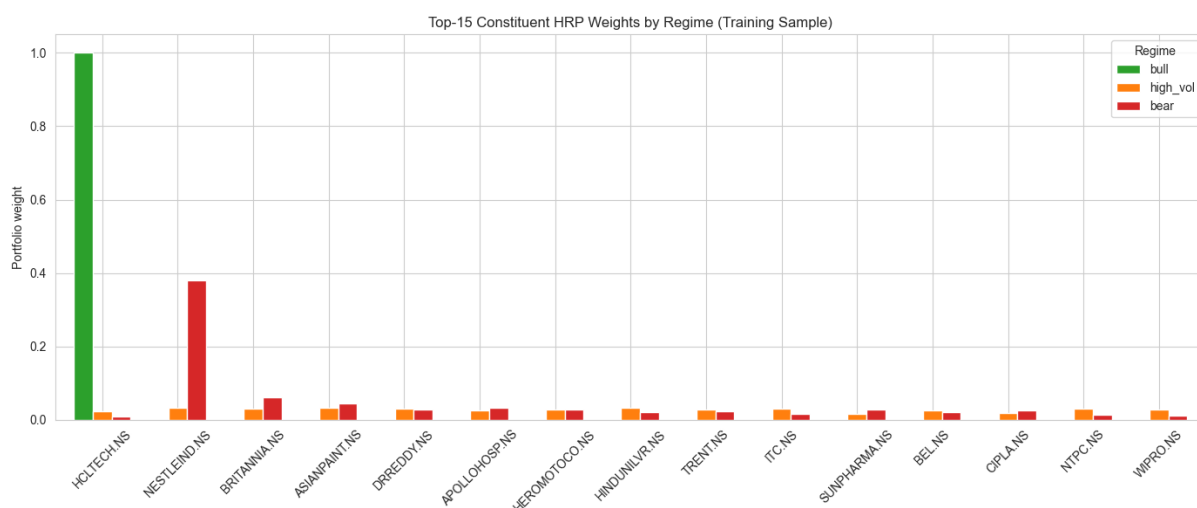


Figure 4. Top-15 constituent HRP weights by regime. The Bull allocation collapses entirely onto HCLTECH (a degenerate single-stock solution), while the Bear and High-Vol portfolios spread across defensive names such as NESTLEIND, BRITANNIA, HINDUNILVR, ASIANPAINT and several pharmaceutical stocks.

Figure 4 shows the leading weights by regime, and the concentration figures in Table 8 summarise them through the Herfindahl–Hirschman Index (HHI). The Bull book is fully concentrated in a single name rather than expressing a deliberate bet, whereas the High-Volatility book is close to equal-weight and the Bear book is only moderately concentrated.

Table 8. Portfolio concentration (Herfindahl–Hirschman Index).

Regime	HHI	Effective N ($1/\text{HHI}$)
Bull	1.0000	1.0 (fully concentrated in HCLTECH)
High-Vol	~ 0.020	~ 49 (near equal-weight)
Bear	~ 0.060	~ 17 (moderately concentrated)

6. Out-of-sample backtest (Jan 2020 – Dec 2024)

The walk-forward backtest rebalances the regime-adaptive HRP strategy at the start of each calendar month, and additionally whenever the HMM signals a regime change that then holds for at least five consecutive trading days (the confirmation buffer). Every rebalance, for all strategies, pays 15 bps of one-way transaction cost. The test window is intentionally unforgiving: it spans the COVID-19 crash (February–April 2020), the liquidity-driven rally of 2020–2021 and the 2022 rate-hike cycle.

Table 9. Full out-of-sample performance summary (risk-free rate 6.5% p.a.).

Strategy	CAGR	Ann. vol	Sharpe	Sortino	Max DD	Calmar	Final wealth
Buy & Hold NIFTY 50	14.19%	19.12%	0.475	0.536	−38.44%	0.37	1.94×
Equal weight (1/N)	25.76%	18.83%	1.002	1.087	−37.58%	0.69	3.14×
Single-regime HRP	26.40%	17.02%	1.120	1.249	−32.43%	0.81	3.22×
Regime-adaptive HRP	38.46%	22.99%	1.285	1.665	−35.31%	1.09	5.08×

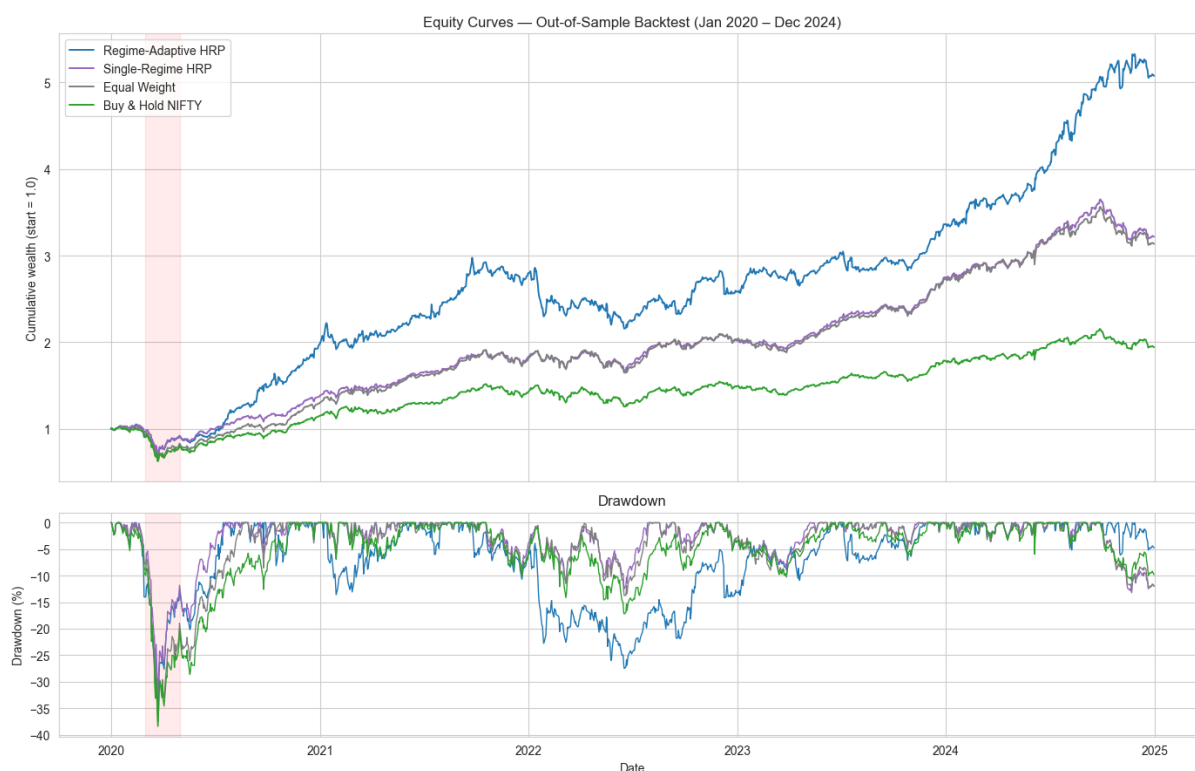


Figure 5. Equity curves (top) and drawdown paths (bottom) for all four strategies over the 2020–2024 test window. The regime-adaptive HRP (blue) reaches the highest terminal wealth ($5.08\times$); its COVID drawdown of -35.31% is shallower than Equal-Weight and Buy & Hold but deeper than Single-Regime HRP.

Table 9 compares the four strategies and Figure 5 plots their equity and drawdown paths. Several results stand out. The regime-adaptive strategy compounds at 38.46% a year, more than double the 14.19% of the passive index over the same five years, and ends the window at $5.08\times$ starting wealth. Its Sharpe ratio of 1.285 sits well above both Buy & Hold (0.475) and the single-regime HRP benchmark (1.120), which indicates that conditioning on the regime adds value beyond plain HRP.

The strategy is not free of risk, however. Its COVID drawdown of -35.31% is shallower than the roughly -37.6% the index gave up over the same crash, but deeper than single-regime HRP (-32.43%), and its annualised volatility of 22.99% is the highest in the table. Both effects trace back to the degenerate Bull weight, which puts 100% into HCLTECH and so amplifies idiosyncratic risk during bull phases. When COVID hit, the model flagged a transition into Bear/High-Vol on 4 March 2020 and, once the five-day buffer cleared on 11 March, rotated into the defensive book. The 62 rebalances over five years (about twelve a year), at an average one-way turnover of 52.7% , are comfortably covered by the 15 bps cost assumption.

7. Robustness and sensitivity

We ran three robustness checks to confirm that the out-of-sample outperformance is not simply an artefact of particular hyperparameter choices.

7.1 Number of HMM states

Table 10 varies the number of states K . With two states the model cannot separate an ordinary bear market from a genuine crisis. With four states a spurious fourth state appears, carrying an extreme -163% annualised return at just 1.1% frequency. Three states minimise the BIC and

give the cleanest economic reading, which is why we adopt $K = 3$ throughout.

Table 10. Sensitivity to the number of HMM states.

States (K)	Bull ann. ret.	Bear ann. ret.	High-Vol ann. ret.	Note
$K = 2$	+12.98%	−8.54%	—	Cannot separate bear from crisis.
$K = 3$	+15.31%	−4.85%	+7.89%	Optimal; minimises the BIC.
$K = 4$	+14.00%	−0.79%	varies	Fourth state has an extreme −163% annualised return at 1.1% frequency.

7.2 Transaction costs

Performance degrades gracefully as costs rise (Table 11). Even at 50 bps, three times the baseline assumption, the strategy still delivers 32.32% CAGR at a Sharpe of 1.08, comfortably ahead of Buy & Hold.

Table 11. Sensitivity to transaction costs.

Transaction cost	CAGR	Sharpe	Sortino	Max DD	Final wealth
5 bps	40.26%	1.342	1.738	−35.18%	5.42×
15 bps	38.46%	1.285	1.665	−35.31%	5.08×
30 bps	35.80%	1.198	1.555	−35.50%	4.61×
50 bps	32.32%	1.080	1.406	−35.76%	4.05×

7.3 Regime confirmation buffer

Results barely move with the buffer length (Table 12): CAGR stays within 38.13–38.69% and the Sharpe ratio within 1.272–1.291 across buffers from one to ten days. The five-day setting we use is conservative and close to optimal.

Table 12. Sensitivity to the regime confirmation buffer.

Buffer (days)	CAGR	Sharpe	Sortino	Max DD	Final wealth
1	38.69%	1.291	1.672	−35.31%	5.13×
3	38.55%	1.288	1.668	−35.31%	5.10×
5	38.46%	1.285	1.665	−35.31%	5.08×
10	38.13%	1.272	1.638	−35.31%	5.02×

8. Summary of results

Table 13. Summary of all results across the pipeline.

Component	Key result
Data	4,236 trading days (2007–2024); 49 NIFTY 50 constituents. Train 3,000 days, test 1,236 days.
HMM	Three-state model: Bull (+15.3%, vol 14.7%), High-Vol (+7.9%, 15.4%), Bear (−4.9%, 40.5%). Bear self-persistence 97.95% (~49-day spells).
Transition matrix	Diagonal [0.001, 0.008, 0.980]. Only Bear is a stable long-duration regime; Bull and High-Vol alternate rapidly.
GARCH	Global $\alpha + \beta = 0.9952$ (near-IGARCH). Per-regime: Bull 0.9717, High-Vol 0.9732, Bear 0.9428. Regime conditioning reduces persistence.
Event mapping	3,594 transitions detected; match rate 2.1% ($\pm 3d$), 3.2% ($\pm 5d$), 6.6% ($\pm 10d$). Major crises (COVID, GFC, taper tantrum, demonetisation) align.
HRP weights	Bull 100% HCLTECH (degenerate); High-Vol and Bear diversified across the 49 stocks. Ledoit–Wolf shrinkage cuts the covariance condition number.
Backtest	Regime-adaptive HRP: 38.46% CAGR, Sharpe 1.28, max DD −35.31%, final wealth $5.08\times$ versus $1.94\times$ for Buy & Hold.
Robustness	Stable across 5–50 bps costs and 1–10 day buffers. $K = 3$ minimises the BIC; $K = 4$ produces a degenerate extreme-return state.
